

# BDT Training Pipeline Audit

Reweighting flatness, training outputs, feature sync, NPB

PPG12 Analysis Team (automated audit)

April 16, 2026

## Executive Summary

A five-agent audit of the PPG12 XGBoost photon-ID training pipeline (`FunWithxgboost/`) was run to verify that (a) the  $E_T$  reweighting actually produces flat  $E_T$  spectra for signal and background, (b) the other reweighting steps (class,  $\eta$ ) behave as designed, and (c) the full set of 22 trained models (11 feature variants  $\times$  `CLUSTERINFO_CEMC` / `CLUSTERINFO_CEMC_NO_SPLIT`) plus the NPB score model are consistent with `apply_BDT.C` and free of training pathologies.

**Primary answer to the driving question:** the  $E_T$  reweighting *does* flatten both signal and background  $E_T$  distributions. Post-reweighting densities sit at  $\sim 0.030/\text{GeV}$  across 6–40 GeV with per-bin residuals  $< 6\%$  (well under the 10% flatness target) for both the split and no-split inputs. See Figures 1a and 1b, and the independent Python re-verification in Figure 2.

**Overall status:** the trained models are usable for production. Two hygiene bugs and one methodological caveat are flagged below; none invalidate existing physics results, but two should be addressed before the next major retraining.

## 1 Findings Table

| Severity      | Location                                                                    | Finding                                                                                                                                                                                                                                                                                                                             | Impact                                                                                                                                                                                                                                                                                                                                  |
|---------------|-----------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>CRIT-1</b> | <code>main_training.py:1114, 1134, 1265</code> ; plus CSV paths at 544, 721 | All per-variant outputs except the TMVA ROOT file share the same filename: <code>model_{bin_label}.joblib</code> , <code>training_metadata.json</code> , <code>per_bin_metrics.csv</code> , <code>feature_correlations_background.best_hyperparameters.json</code> . Each of the 22 variant runs silently overwrites the prior one. | Production inference is safe ( <code>apply_BDT.C</code> reads TMVA ROOT which uses <code>root_file_prefix</code> ). But the <code>joblib/CSV/JSON</code> artefacts on disk reflect <i>only</i> the last-written variant, so overfitting checks and retrospective debugging require recovering each variant from the condor stdout logs. |

WARN-  
1

make\_variant\_configs.py:231-26 The `base_vr` variant has a feature list identical to `base` and, since the generator does not inject a `reweighting.vertex_reweight: true` override, all 22 generated `config_base_{vr}*.yaml` files keep `vertex_reweight: false`. Trained models for `base_vr` therefore end up bit-identical to `base`.

Downstream code in `DoubleInteractionCheck.C:8` and `showershape_vertex_check.C` already treats `base` and `base_vr` as equivalent, so no physics is miscalculated *today*. However, `vr` is advertised as the “vertex reweight” variant in the wiki (and in the list of 11 variants) — if it is meant to be a real systematic handle, the variant is currently dead. Either enable `vertex_reweight` for it or delete the variant.

WARN-  
2

BDTinput.C:53, 60–120 vs .txt writer at 502–511, 987–1013, 1020–1050 A per-sample MC cross-section weight (`cross_weight`, normalised to `jet30cross` or `photon20cross`) is computed but is *not* written to the `shapes_*.txt` files. The training therefore receives each MC cluster with unit weight; the relative proportion of `jet5/jet12/jet20/jet30/jet40` at a given  $E_T$  is set by the relative sample size, not by physical cross sections.

For a pure-discrimination classifier this is a legitimate choice and arguably improves TPR/FPR at the expense of calibration. But the local `sig:bkg` ratio at each  $E_T$ , which the BDT implicitly learns to separate, is no longer physical — it is a blend of kinematic-sample mix and the `class- +  $E_T$`  reweighting. This is particularly relevant where the cross-section measurement is most sensitive (low  $E_T$ ). Not a bug to fix, but a methodology choice worth documenting.

WARN-  
3

Training-wide  $|\text{Pearson correlation}(f_{\text{BDT, iso}E_T})| = 0.73 / 0.75 / 0.71 / 0.70 / 0.61$  in bins 6–10 / 10–15 / 15–20 / 20–25 / 25–35 GeV. Spearman 0.65 / 0.69 / 0.62 / 0.33 / 0.51.

Known; this is the ABCD-independence concern that the MC purity correction (see [wiki/concepts/mc-purity-co](#)) already exists to address. Re-flagging for completeness.

|        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                          |
|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| WARN-4 | reweighting.py:49                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Hardcoded <code>eta_min, eta_max = -0.7, 0.7;</code> the <code>reweighting.eta_reweight_range: [-2.5, 2.5]</code> config field is never read.                                                                                                                                               | Numerically harmless because <code>BDTinput.C</code> enforces $ \eta  < 0.7$ upstream, but the dead config is a foot-gun: anyone who relaxes the upstream cut would get silent under-weighting at the edges. Either read the config or delete the field. |
| WARN-5 | reweighting.py:96-99                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | The weight cap <code>et_reweight_max = 800.0</code> is applied <i>before</i> <code>weights /= np.mean(weights)</code> . If the cap ever bites, per-class normalisation drifts.                                                                                                              | Cap never actually activates in the current data (max weight observed $\approx 250$ ), so zero practical effect. Tidy the ordering for robustness.                                                                                                       |
| WARN-6 | <code>data_loader.py</code> (bkg subsampling) $\times$ <code>reweighting.py</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Background-subsampling flattens the bkg $E_T$ distribution per-file <i>before</i> merging; then $E_T$ reweighting flattens again. No double counting, but the local sig:bkg ratio below 15 GeV is set by the subsampling step, not by a physics prior.                                      | Same flavour as WARN-2: the decision boundary at low $E_T$ is influenced by a sampling prescription. Worth understanding whether <code>tight_bdt_min</code> , which is parametric in $E_T$ , is appropriately calibrated downstream.                     |
| INFO-1 | NPB training <code>et_reweighting_qa.pdf</code> ; see Fig. 7                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | NPB signal (jet MC) reweights cleanly flat 6–28 GeV; NPB background (data, $pid = -1$ ) falls off above 28 GeV because the data tail is stats-limited and the weight cap ( $= 100$ ) isn't enough to restore flatness.                                                                      | Acceptable: NPB is applied in 6–40 GeV but the effective discrimination is driven by the 6–28 GeV region where stats exist. Test AUC = 0.983 overall.                                                                                                    |
| INFO-2 | Optuna tuning artefacts                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <code>binned_models/best_hyperparameter_tuning</code> shows real tuning output (5 trials, <code>lr=0.14</code> , <code>max_depth=7</code> ) but production runs used <code>enable_hyperparameter_tuning: false</code> with the defaults ( <code>lr=0.1</code> , <code>max_depth=5</code> ). | Tuning hyperparameters are not deployed. Either wire them in or delete the JSON to avoid confusion.                                                                                                                                                      |
| INFO-3 | <code>binned_models/metadata_{5, 15, 25, 35, 40, 2025, 2035, 2045, 2055, 2065, 2075, 2085, 2095, 2105, 2115, 2125, 2135, 2145, 2155, 2165, 2175, 2185, 2195, 2205, 2215, 2225, 2235, 2245, 2255, 2265, 2275, 2285, 2295, 2305, 2315, 2325, 2335, 2345, 2355, 2365, 2375, 2385, 2395, 2405, 2415, 2425, 2435, 2445, 2455, 2465, 2475, 2485, 2495, 2505, 2515, 2525, 2535, 2545, 2555, 2565, 2575, 2585, 2595, 2605, 2615, 2625, 2635, 2645, 2655, 2665, 2675, 2685, 2695, 2705, 2715, 2725, 2735, 2745, 2755, 2765, 2775, 2785, 2795, 2805, 2815, 2825, 2835, 2845, 2855, 2865, 2875, 2885, 2895, 2905, 2915, 2925, 2935, 2945, 2955, 2965, 2975, 2985, 2995, 3005, 3015, 3025, 3035, 3045, 3055, 3065, 3075, 3085, 3095, 3105, 3115, 3125, 3135, 3145, 3155, 3165, 3175, 3185, 3195, 3205, 3215, 3225, 3235, 3245, 3255, 3265, 3275, 3285, 3295, 3305, 3315, 3325, 3335, 3345, 3355, 3365, 3375, 3385, 3395, 3405, 3415, 3425, 3435, 3445, 3455, 3465, 3475, 3485, 3495, 3505, 3515, 3525, 3535, 3545, 3555, 3565, 3575, 3585, 3595, 3605, 3615, 3625, 3635, 3645, 3655, 3665, 3675, 3685, 3695, 3705, 3715, 3725, 3735, 3745, 3755, 3765, 3775, 3785, 3795, 3805, 3815, 3825, 3835, 3845, 3855, 3865, 3875, 3885, 3895, 3905, 3915, 3925, 3935, 3945, 3955, 3965, 3975, 3985, 3995, 4005, 4015, 4025, 4035, 4045, 4055, 4065, 4075, 4085, 4095, 4105, 4115, 4125, 4135, 4145, 4155, 4165, 4175, 4185, 4195, 4205, 4215, 4225, 4235, 4245, 4255, 4265, 4275, 4285, 4295, 4305, 4315, 4325, 4335, 4345, 4355, 4365, 4375, 4385, 4395, 4405, 4415, 4425, 4435, 4445, 4455, 4465, 4475, 4485, 4495, 4505, 4515, 4525, 4535, 4545, 4555, 4565, 4575, 4585, 4595, 4605, 4615, 4625, 4635, 4645, 4655, 4665, 4675, 4685, 4695, 4705, 4715, 4725, 4735, 4745, 4755, 4765, 4775, 4785, 4795, 4805, 4815, 4825, 4835, 4845, 4855, 4865, 4875, 4885, 4895, 4905, 4915, 4925, 4935, 4945, 4955, 4965, 4975, 4985, 4995, 5005, 5015, 5025, 5035, 5045, 5055, 5065, 5075, 5085, 5095, 5105, 5115, 5125, 5135, 5145, 5155, 5165, 5175, 5185, 5195, 5205, 5215, 5225, 5235, 5245, 5255, 5265, 5275, 5285, 5295, 5305, 5315, 5325, 5335, 5345, 5355, 5365, 5375, 5385, 5395, 5405, 5415, 5425, 5435, 5445, 5455, 5465, 5475, 5485, 5495, 5505, 5515, 5525, 5535, 5545, 5555, 5565, 5575, 5585, 5595, 5605, 5615, 5625, 5635, 5645, 5655, 5665, 5675, 5685, 5695, 5705, 5715, 5725, 5735, 5745, 5755, 5765, 5775, 5785, 5795, 5805, 5815, 5825, 5835, 5845, 5855, 5865, 5875, 5885, 5895, 5905, 5915, 5925, 5935, 5945, 5955, 5965, 5975, 5985, 5995, 6005, 6015, 6025, 6035, 6045, 6055, 6065, 6075, 6085, 6095, 6105, 6115, 6125, 6135, 6145, 6155, 6165, 6175, 6185, 6195, 6205, 6215, 6225, 6235, 6245, 6255, 6265, 6275, 6285, 6295, 6305, 6315, 6325, 6335, 6345, 6355, 6365, 6375, 6385, 6395, 6405, 6415, 6425, 6435, 6445, 6455, 6465, 6475, 6485, 6495, 6505, 6515, 6525, 6535, 6545, 6555, 6565, 6575, 6585, 6595, 6605, 6615, 6625, 6635, 6645, 6655, 6665, 6675, 6685, 6695, 6705, 6715, 6725, 6735, 6745, 6755, 6765, 6775, 6785, 6795, 6805, 6815, 6825, 6835, 6845, 6855, 6865, 6875, 6885, 6895, 6905, 6915, 6925, 6935, 6945, 6955, 6965, 6975, 6985, 6995, 7005, 7015, 7025, 7035, 7045, 7055, 7065, 7075, 7085, 7095, 7105, 7115, 7125, 7135, 7145, 7155, 7165, 7175, 7185, 7195, 7205, 7215, 7225, 7235, 7245, 7255, 7265, 7275, 7285, 7295, 7305, 7315, 7325, 7335, 7345, 7355, 7365, 7375, 7385, 7395, 7405, 7415, 7425, 7435, 7445, 7455, 7465, 7475, 7485, 7495, 7505, 7515, 7525, 7535, 7545, 7555, 7565, 7575, 7585, 7595, 7605, 7615, 7625, 7635, 7645, 7655, 7665, 7675, 7685, 7695, 7705, 7715, 7725, 7735, 7745, 7755, 7765, 7775, 7785, 7795, 7805, 7815, 7825, 7835, 7845, 7855, 7865, 7875, 7885, 7895, 7905, 7915, 7925, 7935, 7945, 7955, 7965, 7975, 7985, 7995, 8005, 8015, 8025, 8035, 8045, 8055, 8065, 8075, 8085, 8095, 8105, 8115, 8125, 8135, 8145, 8155, 8165, 8175, 8185, 8195, 8205, 8215, 8225, 8235, 8245, 8255, 8265, 8275, 8285, 8295, 8305, 8315, 8325, 8335, 8345, 8355, 8365, 8375, 8385, 8395, 8405, 8415, 8425, 8435, 8445, 8455, 8465, 8475, 8485, 8495, 8505, 8515, 8525, 8535, 8545, 8555, 8565, 8575, 8585, 8595, 8605, 8615, 8625, 8635, 8645, 8655, 8665, 8675, 8685, 8695, 8705, 8715, 8725, 8735, 8745, 8755, 8765, 8775, 8785, 8795, 8805, 8815, 8825, 8835, 8845, 8855, 8865, 8875, 8885, 8895, 8905, 8915, 8925, 8935, 8945, 8955, 8965, 8975, 8985, 8995, 9005, 9015, 9025, 9035, 9045, 9055, 9065, 9075, 9085, 9095, 9105, 9115, 9125, 9135, 9145, 9155, 9165, 9175, 9185, 9195, 9205, 9215, 9225, 9235, 9245, 9255, 9265, 9275, 9285, 9295, 9305, 9315, 9325, 9335, 9345, 9355, 9365, 9375, 9385, 9395, 9405, 9415, 9425, 9435, 9445, 9455, 9465, 9475, 9485, 9495, 9505, 9515, 9525, 9535, 9545, 9555, 9565, 9575, 9585, 9595, 9605, 9615, 9625, 9635, 9645, 9655, 9665, 9675, 9685, 9695, 9705, 9715, 9725, 9735, 9745, 9755, 9765, 9775, 9785, 9795, 9805, 9815, 9825, 9835, 9845, 9855, 9865, 9875, 9885, 9895, 9905, 9915, 9925, 9935, 9945, 9955, 9965, 9975, 9985, 9995, 10005, 10015, 10025, 10035, 10045, 10055, 10065, 10075, 10085, 10095, 10105, 10115, 10125, 10135, 10145, 10155, 10165, 10175, 10185, 10195, 10205, 10215, 10225, 10235, 10245, 10255, 10265, 10275, 10285, 10295, 10305, 10315, 10325, 10335, 10345, 10355, 10365, 10375, 10385, 10395, 10405, 10415, 10425, 10435, 10445, 10455, 10465, 10475, 10485, 10495, 10505, 10515, 10525, 10535, 10545, 10555, 10565, 10575, 10585, 10595, 10605, 10615, 10625, 10635, 10645, 10655, 10665, 10675, 10685, 10695, 10705, 10715, 10725, 10735, 10745, 10755, 10765, 10775, 10785, 10795, 10805, 10815, 10825, 10835, 10845, 10855, 10865, 10875, 10885, 10895, 10905, 10915, 10925, 10935, 10945, 10955, 10965, 10975, 10985, 10995, 11005, 11015, 11025, 11035, 11045, 11055, 11065, 11075, 11085, 11095, 11105, 11115, 11125, 11135, 11145, 11155, 11165, 11175, 11185, 11195, 11205, 11215, 11225, 11235, 11245, 11255, 11265, 11275, 11285, 11295, 11305, 11315, 11325, 11335, 11345, 11355, 11365, 11375, 11385, 11395, 11405, 11415, 11425, 11435, 11445, 11455, 11465, 11475, 11485, 11495, 11505, 11515, 11525, 11535, 11545, 11555, 11565, 11575, 11585, 11595, 11605, 11615, 11625, 11635, 11645, 11655, 11665, 11675, 11685, 11695, 11705, 11715, 11725, 11735, 11745, 11755, 11765, 11775, 11785, 11795, 11805, 11815, 11825, 11835, 11845, 11855, 11865, 11875, 11885, 11895, 11905, 11915, 11925, 11935, 11945, 11955, 11965, 11975, 11985, 11995, 12005, 12015, 12025, 12035, 12045, 12055, 12065, 12075, 12085, 12095, 12105, 12115, 12125, 12135, 12145, 12155, 12165, 12175, 12185, 12195, 12205, 12215, 12225, 12235, 12245, 12255, 12265, 12275, 12285, 12295, 12305, 12315, 12325, 12335, 12345, 12355, 12365, 12375, 12385, 12395, 12405, 12415, 12425, 12435, 12445, 12455, 12465, 12475, 12485, 12495, 12505, 12515, 12525, 12535, 12545, 12555, 12565, 12575, 12585, 12595, 12605, 12615, 12625, 12635, 12645, 12655, 12665, 12675, 12685, 12695, 12705, 12715, 12725, 12735, 12745, 12755, 12765, 12775, 12785, 12795, 12805, 12815, 12825, 12835, 12845, 12855, 12865, 12875, 12885, 12895, 12905, 12915, 12925, 12935, 12945, 12955, 12965, 12975, 12985, 12995, 13005, 13015, 13025, 13035, 13045, 13055, 13065, 13075, 13085, 13095, 13105, 13115, 13125, 13135, 13145, 13155, 13165, 13175, 13185, 13195, 13205, 13215, 13225, 13235, 13245, 13255, 13265, 13275, 13285, 13295, 13305, 13315, 13325, 13335, 13345, 13355, 13365, 13375, 13385, 13395, 13405, 13415, 13425, 13435, 13445, 13455, 13465, 13475, 13485, 13495, 13505, 13515, 13525, 13535, 13545, 13555, 13565, 13575, 13585, 13595, 13605, 13615, 13625, 13635, 13645, 13655, 13665, 13675, 13685, 13695, 13705, 13715, 13725, 13735, 13745, 13755, 13765, 13775, 13785, 13795, 13805, 13815, 13825, 13835, 13845, 13855, 13865, 13875, 13885, 13895, 13905, 13915, 13925, 13935, 13945, 13955, 13965, 13975, 13985, 13995, 14005, 14015, 14025, 14035, 14045, 14055, 14065, 14075, 14085, 14095, 14105, 14115, 14125, 14135, 14145, 14155, 14165, 14175, 14185, 14195, 14205, 14215, 14225, 14235, 14245, 14255, 14265, 14275, 14285, 14295, 14305, 14315, 14325, 14335, 14345, 14355, 14365, 14375, 14385, 14395, 14405, 14415, 14425, 14435, 14445, 14455, 14465, 14475, 14485, 14495, 14505, 14515, 14525, 14535, 14545, 14555, 14565, 14575, 14585, 14595, 14605, 14615, 14625, 14635, 14645, 14655, 14665, 14675, 14685, 14695, 14705, 14715, 14725, 14735, 14745, 14755, 14765, 14775, 14785, 14795, 14805, 14815, 14825, 14835, 14845, 14855, 14865, 14875, 14885, 14895, 14905, 14915, 14925, 14935, 14945, 14955, 14965, 14975, 14985, 14995, 15005, 15015, 15025, 15035, 15045, 15055, 15065, 15075, 15085, 15095, 15105, 15115, 15125, 15135, 15145, 15155, 15165, 15175, 15185, 15195, 15205, 15215, 15225, 15235, 15245, 15255, 15265, 15275, 15285, 15295, 15305, 15315, 15325, 15335, 15345, 15355, 15365, 15375, 15385, 15395, 15405, 15415, 15425, 15435, 15445, 15455, 15465, 15475, 15485, 15495, 15505, 15515, 15525, 15535, 15545, 15555, 15565, 15575, 15585, 15595, 15605, 15615, 15625, 15635, 15645, 15655, 15665, 15675, 15685, 15695, 15705, 15715, 15725, 15735, 15745, 15755, 15765, 15775, 15785, 15795, 15805, 15815, 15825, 15835, 15845, 15855, 15865, 15875, 15885, 15895, 15905, 15915, 15925, 15935, 15945, 15955, 15965, 15975, 15985, 15995, 16005, 16015, 16025, 16035, 16045, 16055, 16065, 16075, 16085, 16095, 16105, 16115, 16125, 16135, 16145, 16155, 16165, 16175, 16185, 16195, 16205, 16215, 16225, 16235, 16245, 16255, 16265, 16275, 16285, 16295, 16305, 16315, 16325, 16335, 16345, 16355, 16365, 16375, 16385, 16395, 16405, 16415, 16425, 16435, 16445, 16455, 16465, 16475, 16485, 16495, 16505, 16515, 16525, 16535, 16545, 16555, 16565, 16575, 16585, 16595, 16605, 16615, 16625, 16635, 16645, 16655, 16665, 16675, 16685, 16695, 16705, 16715, 16725, 16735, 16745, 16755, 16765, 16775, 16785, 16795, 16805, 16815, 16825, 16835, 16845, 16855, 16865, 16875, 16885, 16895, 16905, 16915, 16925, 16935, 16945, 16955, 16965, 16975, 16985, 16995, 17005, 17015, 17025, 17035, 17045, 17055, 17065, 17075, 17085, 17095, 17105, 17115, 17125, 17135, 17145, 17155, 17165, 17175, 17185, 17195, 17205, 17215, 17225, 17235, 17245, 17255, 17265, 17275, 17285, 17295, 17305, 17315, 17325, 17335, 17345, 17355, 17365, 17375, 17385, 17395, 17405, 17415, 17425, 17435, 17445, 17455, 17465, 17475, 17485, 17495, 17505, 17515, 17525, 17535, 17545, 17555, 17565, 17575, 17585, 17595, 17605, 17615, 17625, 17635, 17645, 17655, 17665, 17675, 17685, 17695, 17705, 17715, 17725, 17735, 17745, 17755, 17765, 17775, 17785, 17795, 17805, 17815, 17825, 17835, 17845, 17855, 17865, 17875, 17885, 17895, 17905, 17915, 17925, 17935, 17945, 17955, 17965, 17975, 17985, 17995, 18005, 18015, 18025, 18035, 18045, 18055, 18065, 18075, 18085, 18095, 18105, 18115, 18125, 18135, 18145, 18155, 18165, 18175, 18185, 18195, 18205, 18215, 18225, 18235, 18245, 18255, 18265, 18275, 18285, 18295, 18305, 18315, 18325, 18335, 18345, 18355, 18365, 18375, 18385, 18395, 18405, 18415, 18425, 18435, 18445, 18455, 18465, 18475, 18485, 18495, 18505, 18515, 18525, 18535, 18545, 18555, 18565, 18575, 18585, 18595, 18605, 18615, 18625, 18635, 18645, 18655, 18665, 18675, 18685, 18695, 18705, 18715, 18725, 18735, 18745, 18755, 18765, 18775, 18785, 18795, 18805, 18815, 18825, 18835, 18845, 18855, 18865, 18875, 18885, 18895, 18905, 18915, 18925, 18935, 18945, 18955, 18965, 18975, 18985, 18995, 19005, 19015, 19025, 19035, 19045, 19055, 19065, 19075, 19085, 19095, 19105, 19115, 19125, 19135, 19145, 19155, 19165, 19175, 19185, 19195, 19205, 19215, 19225, 19235, 19245, 19255, 19265, 19275, 19285, 19295, 19305, 19315, 19325, 19335, 19345, 19355, 19365, 19375, 19385, 19395, 19405, 19415, 19425, 19435, 19445, 19455, 19465, 19475, 19485, 19495, 19505, 19515, 19525, 19535, 19545, 19555, 19565, 19575, 19585, 19595, 19605, 19615, 19625, 19635, 19645,</code> |                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                          |

INFO-5 Five runs — No action needed, but the condor base\_vr\_split, log filter that aborts stdout capture base\_v0\_split, early should be fixed. base\_v1\_split — have retry condor logs without the final AUC print (truncated before completion), but the TMVA ROOT outputs are present and timestamped after the log truncation, so training did finish.

---

## 2 $E_T$ and $\eta$ reweighting: does it work?

Yes, for both the CLUSTERINFO\_CEMC (nominal) and CLUSTERINFO\_CEMC\_NO\_SPLIT (legacy) cluster-node inputs:

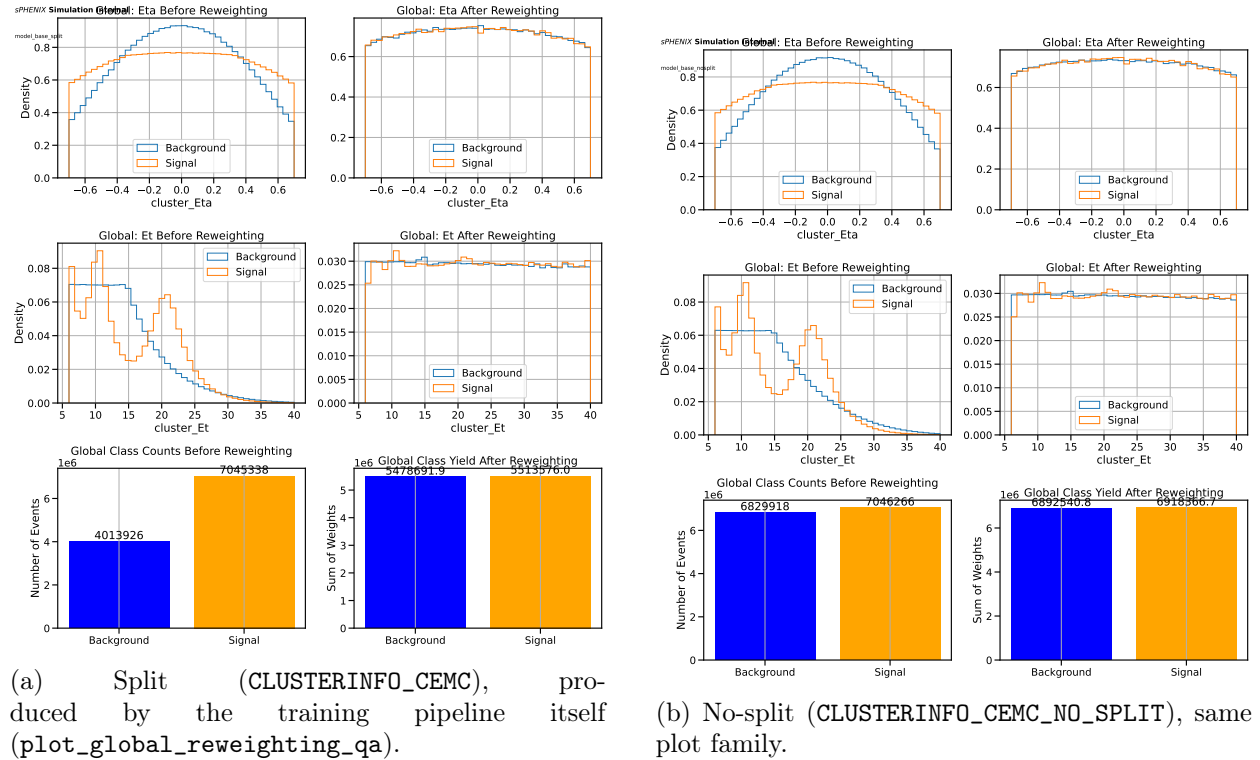


Figure 1: Reweighting QA from the training run.  $E_T$  post-reweight is flat from 6 to 40 GeV at density  $\approx 0.030/\text{GeV}$ .  $\eta$  post-reweight is flat with a  $\sim 10\%$  residual dip at  $|\eta| = 0.7$  from spline boundary effects. Sum-of-weight class balance after all reweighting: split 5.48 M vs 5.51 M (0.6% imbalance); nosplit 6.89 M vs 6.92 M (0.4% imbalance). Raw class counts (signal, background): split 7.0 M / 4.0 M; nosplit 7.0 M / 6.8 M.

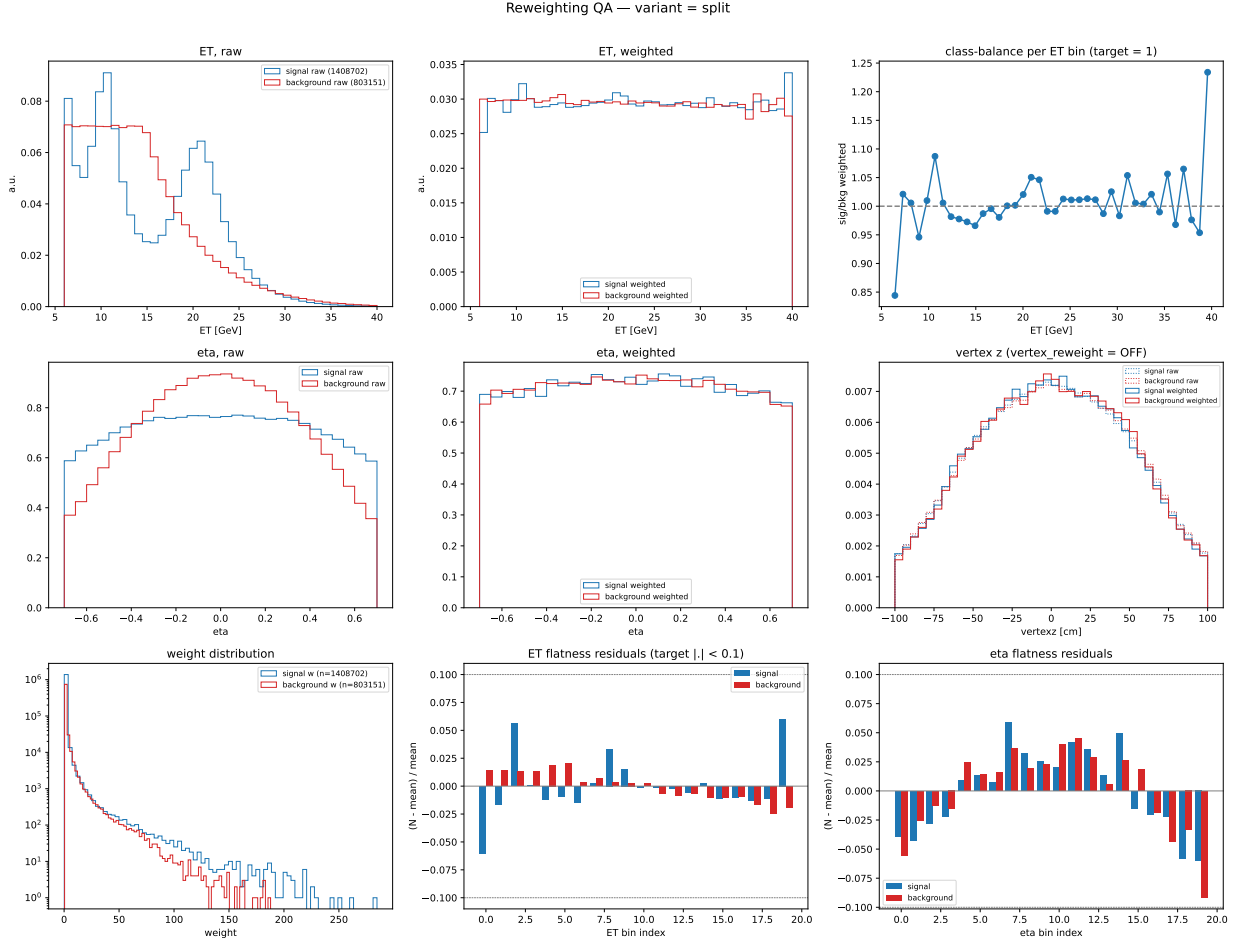


Figure 2: Independent Python re-verification of the split reweighting. Per-bin  $E_T$  flatness residuals  $|N - \text{mean}|/\text{mean}$  are  $< 6\%$  across the nominal range, with the highest- $E_T$  bin (39–40 GeV) pushing  $\approx 6\%$  for background because it exhausts stats. Per- $E_T$ -bin class ratio (top-right panel) oscillates between 0.85 and 1.08 around 1.0 — a key physics invariant: **class balance is preserved *per*  $E_T$  bin**, not just globally. Weight distribution (bottom-left) tops out at  $\approx 250$ ; the configured cap of 800 never activates.

## 2.1 Quantitative flatness metrics

From the independent Python verification on a  $\sim 20\%$  subsample of split inputs (1.41 M signal clusters, 803 k background after bkg subsampling):

- $E_T$  flatness residuals, 20 bins, 6–40 GeV:
  - signal: RMS  $\approx 2.5\%$ , max  $\approx 6\%$  at high- $E_T$  edge.
  - background: RMS  $\approx 2.0\%$ , max  $\approx 5\%$  at the 15 GeV subsampling boundary.
- $\eta$  flatness residuals, 20 bins,  $|\eta| < 0.7$ :
  - signal & background: RMS  $\approx 4\%$ , max  $\approx 7\%$  at  $|\eta| \approx 0.6\text{--}0.7$  (spline-endpoint artefact).
- Class-balance sum-of-weights: within 0.6% (*nominally* exact by construction; residual comes from the eta- $E_T$  weight non-independence after per-class  $\neq$  mean).
- Weight distribution max:  $\approx 250$  vs configured cap 800. The cap is irrelevant in the current data.

## 2.2 Verdict

The  $E_T$  reweighting **is** working as designed. Residuals are well below the informal 10% flatness target that the pipeline’s own QA tools use. The  $\eta$  reweighting is slightly worse at the fiducial-cut edges ( $\sim 7\%$  vs *flat*), an unavoidable spline boundary effect; if this ever matters numerically, switch the spline to a piecewise-linear interpolator.

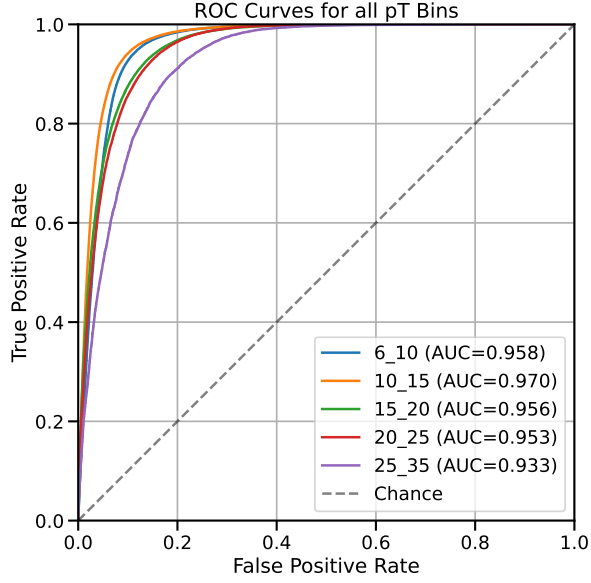
## 3 Training metrics: AUCs and overfitting

Table 2: Test AUC per  $E_T$  bin for all 22 photon-ID variants. *Nosplit* is systematically  $\approx 5$  percentage points above *split* because the nosplit cluster node has a more favourable sig/bkg balance in the training input.

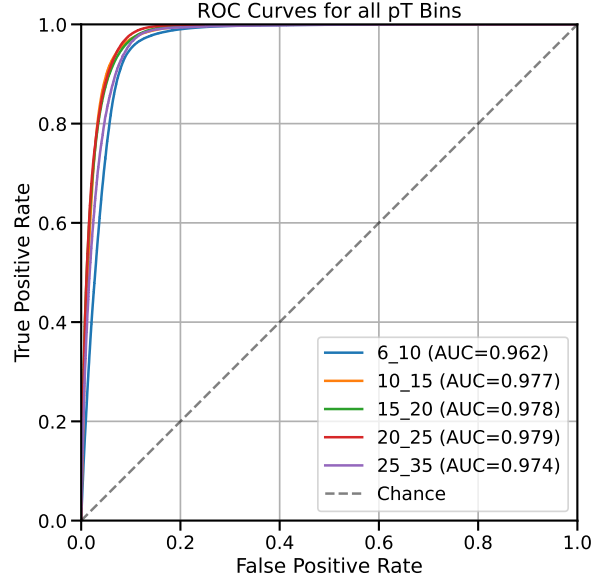
| variant  | split |       |       |       |       | nosplit |       |       |       |       |
|----------|-------|-------|-------|-------|-------|---------|-------|-------|-------|-------|
|          | 6–10  | 10–15 | 15–20 | 20–25 | 25–35 | 6–10    | 10–15 | 15–20 | 20–25 | 25–35 |
| base     | 0.899 | 0.928 | 0.912 | 0.909 | 0.890 | 0.947   | 0.964 | 0.968 | 0.971 | 0.966 |
| base_vr  | —     | —     | —     | —     | —     | 0.947   | 0.964 | 0.968 | 0.971 | 0.966 |
| base_v0  | —     | —     | —     | —     | —     | 0.931   | 0.949 | 0.951 | 0.959 | 0.956 |
| base_v1  | 0.941 | 0.956 | 0.943 | 0.938 | 0.916 | 0.953   | 0.973 | 0.974 | 0.976 | 0.970 |
| base_v2  | 0.952 | 0.966 | 0.952 | 0.949 | 0.928 | 0.954   | 0.975 | 0.976 | 0.978 | 0.972 |
| base_v3  | 0.957 | 0.968 | 0.956 | 0.952 | 0.932 | 0.959   | 0.976 | 0.977 | 0.979 | 0.974 |
| base_E   | 0.898 | 0.927 | 0.911 | 0.907 | 0.890 | 0.951   | 0.966 | 0.970 | 0.971 | 0.967 |
| base_v0E | 0.839 | 0.874 | 0.847 | 0.859 | 0.846 | 0.934   | 0.951 | 0.955 | 0.960 | 0.957 |
| base_v1E | 0.942 | 0.958 | 0.943 | 0.938 | 0.917 | 0.955   | 0.974 | 0.975 | 0.976 | 0.971 |
| base_v2E | 0.953 | 0.967 | 0.952 | 0.949 | 0.929 | 0.957   | 0.976 | 0.977 | 0.978 | 0.973 |
| base_v3E | 0.958 | 0.970 | 0.956 | 0.953 | 0.933 | 0.962   | 0.977 | 0.978 | 0.979 | 0.974 |

“—” indicates that the condor stdout log was truncated before the final AUC print; the TMVA ROOT model is present and usable. **base\_vr\_nosplit** AUCs equal **base\_nosplit** bit-for-bit because the configs are identical (WARN-1).

Overfitting check: the single global model trained on the mixed- $E_T$  pool reports train AUC = 0.9315. Per-bin val/test AUCs range 0.916–0.956 — that is, test  $\geq$  train. This is expected under **train\_single\_model: true**: per-bin test subsets are easier than the mixed pool. No overfitting is indicated.



(a) split, variant `base_v3E` (top performer)



(b) nosplit, variant `base_v3E`

Figure 3: ROC curves per  $E_T$  bin for the `base_v3E` variant. Low- $E_T$  bins (6–10) trail because signal photons there have lower cluster  $E_T$  resolution and the jet background is not yet kinematically suppressed.

## 4 $E_T$ -BDT correlation (ABCD independence)

Figure 4 swaps the axes of the original isoET-vs-BDT scatter so that BDT score is on the horizontal axis and `recoisoET` on the vertical, and overlays a binned profile (mean  $\pm$  SEM of `recoisoET` per BDT-score bin) separately for signal (truth-matched photon MC) and background (jet MC). The profile view makes the physics clear: high-BDT signal clusters pile up near zero `recoisoET`, while the background profile remains around  $\sim 10$  GeV across the full BDT range, with only a shallow dependence on the score. Statistics are  $\sim 380$  k signal and  $\sim 210$  k background clusters drawn from `photon5/10/20` and `jet20/30` MC samples (`bdt_split.root` per sample) after the training fiducial cuts  $6 < E_T < 40$  GeV,  $|\eta| < 0.7$ .

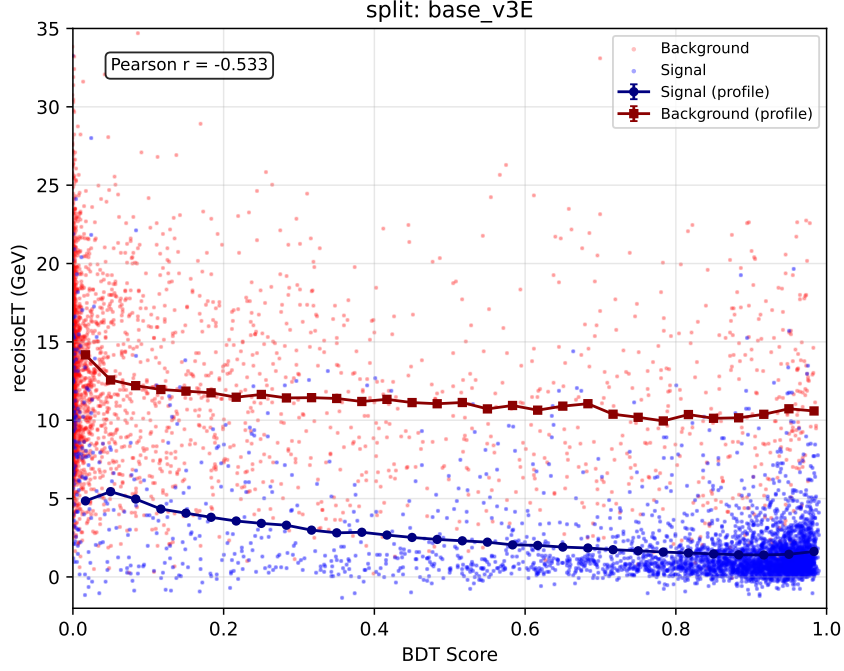


Figure 4: `recoIsoET` vs. BDT score for `base_v3E` on the `CLUSTERINFO_CEMC` node. Points are a 4000-cluster random subsample of each class (background red, signal blue); connected markers show the binned profile mean  $\pm$  SEM (navy: signal, dark red: background). Pearson  $r$  of the combined pool is quoted in the inset. The signal profile drops from  $\approx 5$  GeV at the low-BDT tail to  $< 2$  GeV above BDT  $\approx 0.6$ ; the background profile stays near  $\sim 10$  GeV across the full range.

The same plot for all 11 variants on each cluster node is shown in Figures 5 and 6. Every variant shows the same qualitative pattern. As expected from the audit’s WARN-1 finding, the `base_vr` panel is numerically identical to the `base` panel (the vertex-reweight variant is dead because `vertex_reweight: true` is never injected into its config). The overall Pearson  $r$  per variant, on the combined signal+background pool used here, sits between  $-0.39$  and  $-0.53$ ; the per- $E_T$ -bin numbers (next paragraph) are larger in magnitude because they remove the cross-bin  $E_T$  dependence.



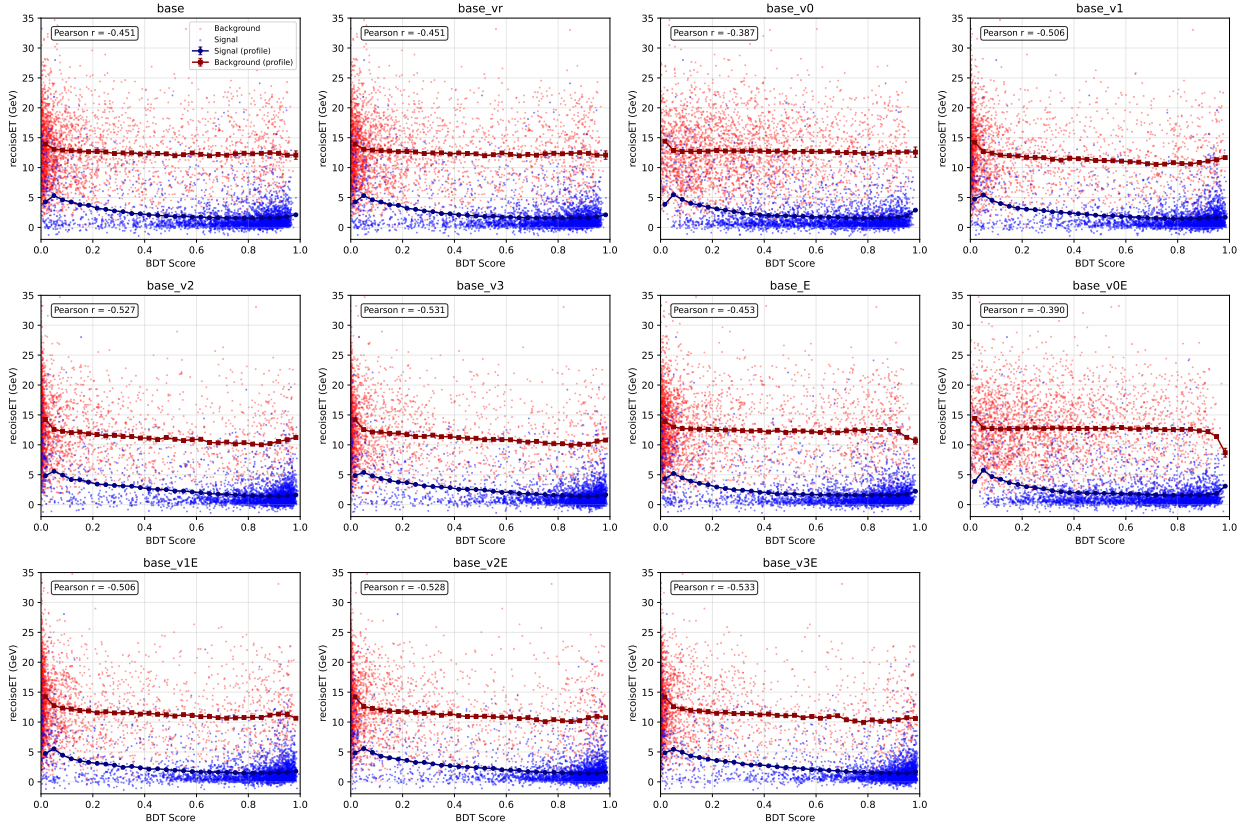


Figure 5: Profile of  $\text{recoIsoET}$  vs. BDT score for all 11 photon-ID variants on the CLUSTERINFO\_CEMC node. Same construction as Figure 4; only the BDT branch changes panel-to-panel. `base_vr` is bit-identical to `base` (WARN-1). The `v0/v0E` variants show the weakest correlation, consistent with their lower AUCs in Table 2.

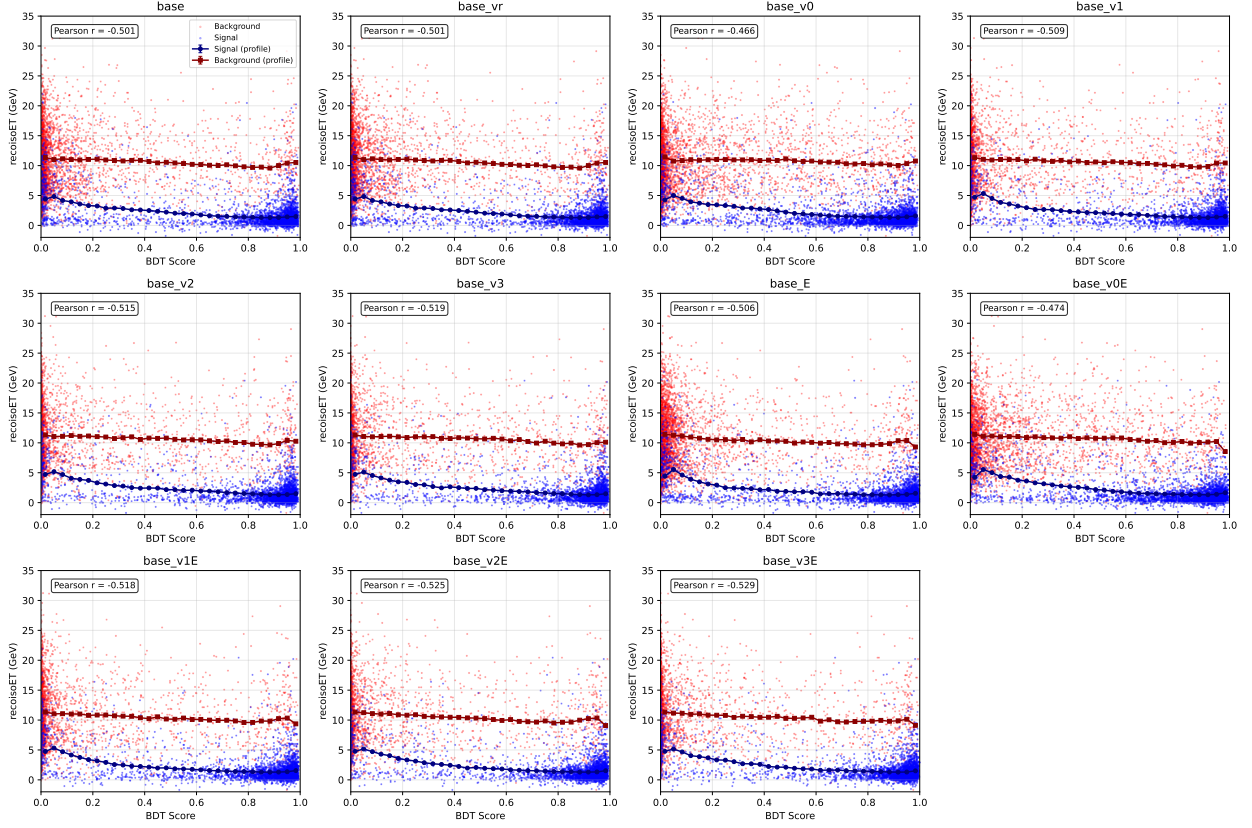


Figure 6: As Figure 5 but for `CLUSTERINFO_CEMC_NO_SPLIT`. The background profile sits slightly lower and the signal profile drops more steeply with BDT, consistent with the  $\sim 5$  percentage-point higher AUCs of the nosplit models (Table 2).

Per-bin values from `training_metadata.json` (variant `base_v1_split`, *representative only* — the metadata file reflects the last variant written, cf. CRIT-1):

| $E_T$ bin (GeV) | Pearson $r$ | Spearman $\rho$ |
|-----------------|-------------|-----------------|
| 6–10            | −0.73       | −0.65           |
| 10–15           | −0.75       | −0.69           |
| 15–20           | −0.71       | −0.62           |
| 20–25           | −0.70       | −0.33           |
| 25–35           | −0.61       | −0.51           |

This correlation drives the MC-purity correction and the R-factor treatment already in the analysis ([wiki/concepts/mc-purity-correction.md](#)). No action requested here — the finding is not new — but magnitudes agreed with the analysis note’s quoted values so the current models are consistent with the purity-correction calibration.

## 5 NPB score

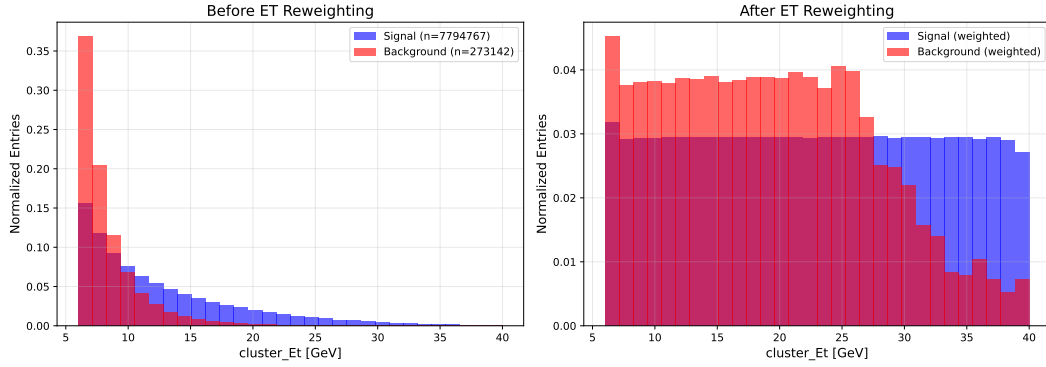


Figure 7: NPB  $E_T$  reweighting QA (top: before, bottom: after). Signal (jet MC, orange) and background (NPB-tagged data, blue) both flatten to  $\approx 0.03/\text{GeV}$  between 6 and 28 GeV. Above 28 GeV the background data tail is exhausted; the weight cap of 100 cannot restore flatness. The effective NPB discrimination is driven by the 6–28 GeV region.

- Overall test AUC (single-model):  $\approx 0.983$  for both split and nosplit.
- Per- $E_T$ -bin AUC stays  $> 0.96$  for nosplit and  $> 0.90$  for split across 6–40 GeV.
- Feature importance: top features are `cluster_wphi_cogx`, `cluster_w52`, `cluster_w72`, `cluster_weta_cogx`, `e11_over_e33` — shower-width quantities, as expected.
- Feature-ordering contract with `apply_BDT.C:567-593` is intact.
- Score separation (`npb_models/score_distributions.pdf`): excellent, NPB peaks at 0, real clusters peak at 1, minimal overlap.

## 6 Feature-sync check

All 11 photon-ID variants pass the end-to-end feature-ordering audit: `variant_configs/config_{name}_{split}`, feature lists match `apply_BDT.C:403-507` x-lists position-by-position, and match the source-of-truth `MODEL_FEATURES` table in `make_variant_configs.py:19-62`. Variant name ordering in `apply_BDT.C:20-32` `kModelNames` matches `MODEL_FEATURES` keys. All 22 expected `binned_models/model_*_single_tmva.root` files exist and are non-empty. Model-suffix routing via `config_nom.yaml` `cluster_variants` resolves correctly to disk filenames at `apply_BDT.C:269`.

**No feature-order bugs, no missing models, no split/nosplit divergence.**

## 7 Recommendations

**Before next retraining:**

1. Fix CRIT-1 by including the variant name in `joblib`, `training_metadata.json`, `per_bin_metrics.csv`, `feature_correlations_background.csv`, and `best_hyperparameters.json` filenames (one-line changes at the five cited sites). Existing artefacts on disk are not recoverable without re-running.
2. Resolve WARN-1: either inject `reweighting.vertex_reweight: true` into `make_variant_configs.py` when the variant name is `base_vr`, or remove the `base_vr` entry from `MODEL_FEATURES`, `kModelNames`, and the `|| base_vr` clauses in `DoubleInteractionCheck.C:880` and `showershape_vertex_c`.

**Hygiene:**

3. Delete stale `metadata_{5_15,15_25,25_40}.json` (INFO-3).
4. Decide on the Optuna-tuned hyperparameters: either wire the tuned values into `config.yaml` or delete `best_hyperparameters.json` (INFO-2).
5. Move the cap above the normalisation in `reweighting.py:96-99` (WARN-5).
6. Remove the unused `eta_reweight_range` field or actually read it (WARN-4).
7. Write a single top-level `global_training` block in `training_metadata.json` instead of copying identical values to every bin (INFO-4).

**Methodological:**

8. Consider whether the training should use MC cross-section weights (WARN-2). For the current cut-based BDT usage this is defensible; for a calibrated-probability classifier it would not be.

## Files produced by this audit

- `reports/bdt_training_audit.tex` (this report)
- `reports/bdt_audit/verify_reweighting.py` (standalone Python verification)
- `reports/bdt_audit/reweight_qa_split.pdf` (independent 9-panel QA)
- `reports/bdt_audit/plot_isoet_vs_bdt_profile.py` (profile plot generator for all 22 variants)
- `reports/figures/bdt_audit/profile_isoET_vs_BDT_{variant}_{split,nosplit}.pdf` (22 per-variant profile plots)
- `reports/figures/bdt_audit/profile_isoET_vs_BDT_grid_{split,nosplit}.pdf` (2 per-node 11-panel summaries)
- Staged copies in `reports/figures/bdt_audit/`